EC5003: APPLIED ECONOMETRICS

LECTURE 11

Multicollinearity

Reading: Wooldridge Chp 3;Pedace Chp 10

Overview

- The definition of multicollinearity (M/C) and the distinction between perfect and imperfect (high) M/C
- The symptoms of M/C
- The detection of M/C
- Solutions to the existence of M/C

Multicollinearity

- An assumption of OLS
  - no exact linear relationship exists between sample values of two/more explanatory variables
  - If there is then OLS will not be feasible

- But a problem that frequently occurs in multiple regression is very near (or high) linear relationships between explanatory variables.

- Arises in <u>multiple</u> regression when movements in one explanatory variable are closely matched by movements in one or more other variables.

- **<u>Consequence</u>** - may not be possible to estimate effectively (ie coefficient standard errors become large) the separate effects of the explanatory variables.

Multicollinearity of Classical Assumptions

- Multicollinearity does not violate any of the key assumptions – so not well defined

- What we hope for when we construct a data set is that there is less correlation between the explanatory variables than more

- **Example** – examine relationship between types of school expenditure and student performance – likely types of expenditure positively correlated – so identifying importance any single category becomes difficult – so either aggregate some of the data or collect more data…

Perfect Multicollinearity

- Population equation

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i$$

- Estimate with a sample of data – but assume

$$X_{2i} = 2 + 3X_{3i}$$

- Sample values X_2 - **<u>exactly linearly related</u>** X_3
- Highly unlikely in practice unless you make a mistake when organising data

- Assume estimate sample regression equation

$$\hat{Y} = 10 + 20\,X_2 + 5\,X_3$$

- 10, 20, and 5 are OLS parameter estimates β_i (i=1,2,3)

- But - linear combination of X_2 and X_3 such that we can rewrite the above equation using $X_{2i} = 2 + 3X_{3i}$:

- Start (A)

$$\hat{Y} = 10 + 10\,X_2 + 10\,X_2 + 5\,X_3$$

$$\hat{Y} = 10 + 10\,X_2 + 10(2 + 3\,X_3) + 5\,X_3$$

- Attain (B)

$$\hat{y} = 30 + 10X_2 + 35X_3$$

Which is Correct? Equation A or B?

- Alternatively

$$\hat{Y} = 10 + 40X_2 - 20X_2 + 5X_3$$

$$\hat{Y} = 10 + 40X_2 - 20(2 + 3X_3) + 5X_3$$

$$\hat{Y} = -30 + 40X_2 - 55X_3$$

- There are an infinite number of possibilities – only happens if **perfect multicollinearity.**

- As $X_2 = 2 + 3X_3$, predicted Y will be the same for all equations.

- So still able to forecast Y if able to estimate the best fit line (but can't estimate if perfect M/C).

- In practice perfect multicollinearity is very rare.
- It might arise if you are unaware of some identity or relationship which holds in your explanatory data (can happen with certain macro aggregates).
- Or it might be the case that there is a common trend in the data.
- More likely to occur because of a data collection mistake (due to sample issues) or data manipulation mistake by the analyst (variables in many data sets are composites of others collected).

Model Specifications

- Polynomial terms can induce M/C although most likely to be imperfect or high.

- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_1^3 + \varepsilon$

- It is quite common practice to introduce polynomial terms into a regression if the relationship between X and Y is assumed to be non-linear (later lecture).

- E.g., Age and earnings.

- It will also arise if you fall into the *Dummy Variable Trap*

- Suppose include 4 seasonal dummies and a constant in a quarterly data regression model.

- $Y_t = \beta_0 + \beta_1 d_{1t} + \beta_2 d_{2t} + \beta_3 d_{3t} + \beta_4 d_{4t} + \varepsilon_t$

where

when $d_1 = 1$ and d_2, d_3 and $d_4 = 0 \rightarrow Y_t = \beta_0 + \beta_1 1$

when $d_2 = 1$ and d_1, d_3 and $d_4 = 0 \rightarrow Y_t = \beta_0 + \beta_2 1$

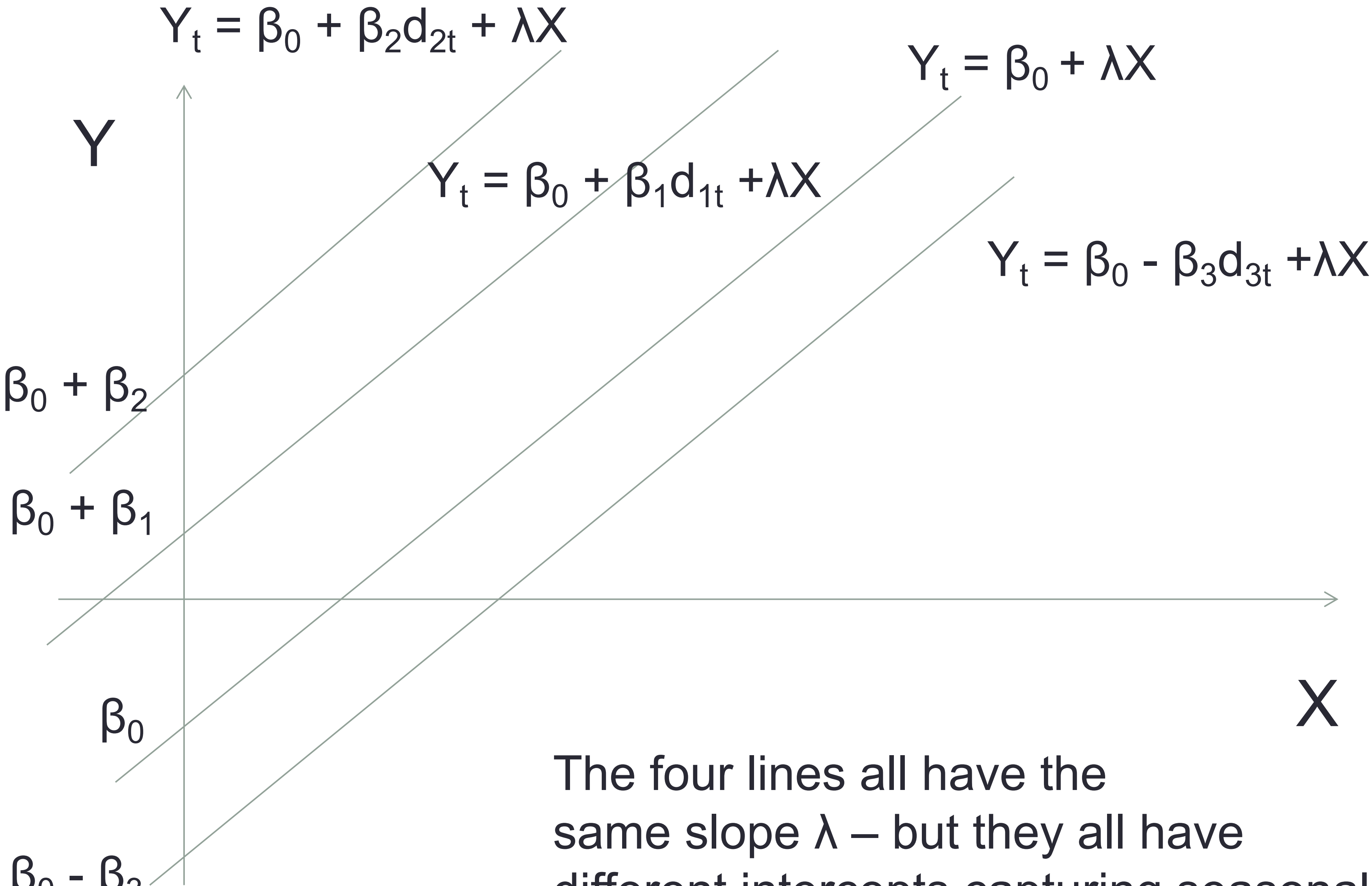
when $d_3 = 1$ and d_1, d_2 and $d_4 = 0 \rightarrow Y_t = \beta_0 + \beta_3 1$

when $d_4 = 1$ and d_1, d_2 and $d_3 = 0 \rightarrow Y_t = \beta_0 + \beta_4 1$

- Each dummy variable is equal to one, which is equivalent to the constant (β_0).
- So we have two constants in the regression, or exact linear dependence in the data matrix.
- So we drop one of the dummy variables (e.g., β_4).
- $Y_t = \beta_0 + \beta_1 d_{1t} + \beta_2 d_{2t} + \beta_3 d_{3t} + \varepsilon_t$
- This is then referred to as the base quarter (see next slide)
- We interpret $\beta_{1,2,3}$ relative to the base quarter which is captured by β_0

So if we include all dummy variables unable to solve model

Computer will say no!



The four lines all have the same slope λ – but they all have different intercepts capturing seasonal variation in the relationship

Imperfect (Very High) M/C

- Relationship between variables now a slight modification to the perfect linear example.

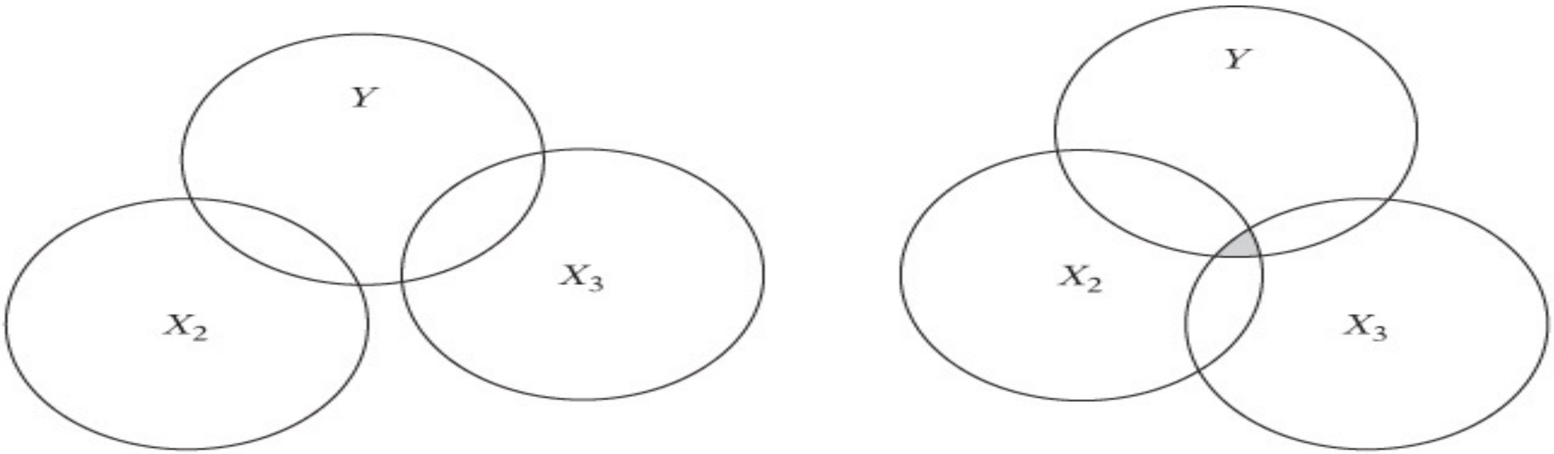
- The variables are now less than perfectly correlated due to the presence of a random error.

- So the linear relationship between the variables is almost but not quite exact.

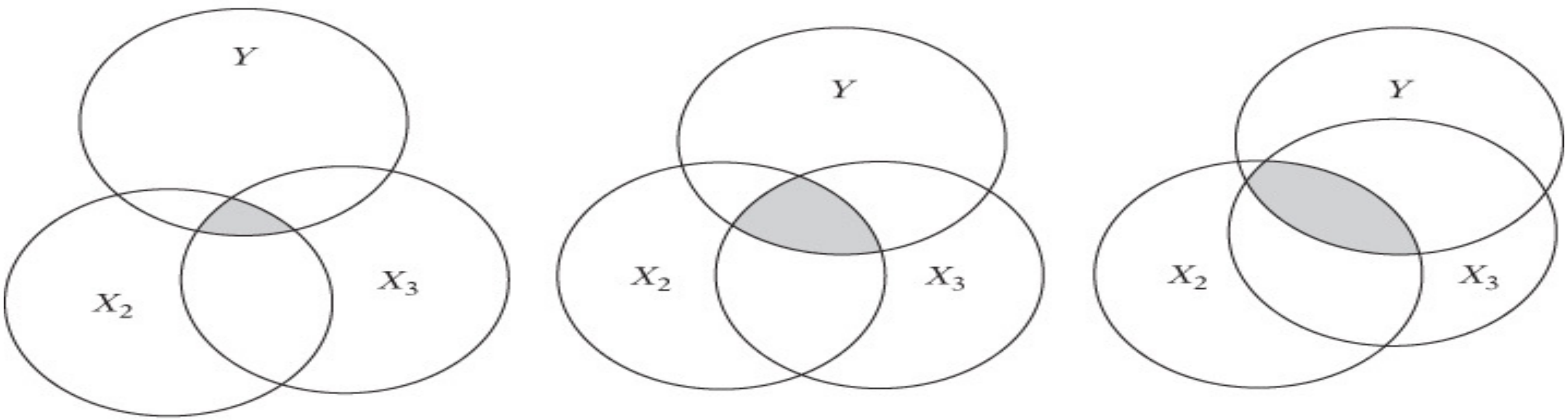
- $X_{2i} = 2 + 3X_{3i} + \varepsilon_i$ (this is a random error).

- This problem is endemic with economic data, particularly time series data with common trends, and autoregressive-distributed lag dynamic models
- In this case able to estimate exact estimates for β_i
- But there will be many sets of regressions results for which the residual sum of squares (RSS) are similar.
- The existence of high MC will manifest in the degree of statistical precision – relative size of standard errors.
- Large standard errors – problem with inference especially T-tests for individual parameter estimates.
- Insignificant t-statistics

M/C – A Picture



(*a*) No collinearity (*b*) Low collinearity



(*c*) Moderate collinearity (*d*) High collinearity (*e*) Very high collinearity

Consequences and Symptoms (i)

- OLS estimates are still BLUE but, variances and co-variances are likely to be large

- Difficult to achieve precise coefficient estimates

- Statistical inference will be problematic with:
  - wide confidence intervals
  - statistical insignificance likely (i.e. low t-ratios)
  - but R^2 will be high (results look “strange”)

- So there might be an obvious inconsistency in the pattern of your results

Consequences and Symptoms (ii)

- Hence, regression equation appears to explain the dependent variable well but no/several individual explanatory variables appears significant.

- Results can be sensitive to small changes in the sample.

- So if add/delete a few observations in a sample can see large changes in significance of parameter estimates.

Why?

- The inclusion of a few additional data points can seriously influence the position/properties of the best fit line

- Standard Errors - affected - T test TS value low - unable reject null hypothesis:

$$TS = \frac{\hat{\beta} - \beta}{S_{\ddot{\beta}}}$$

- Estimated parameters subject to error.

- Also when M/C present need to be aware of mis-specification error.

- i.e., reject a variable with a high standard error when it should be included as an explanatory variable.

- But residuals not affected – so F Test reliable (R^2 ok).

Applied Example (Consumption Function)

- C = f(Y,L)

- Consumer Expenditure - C
- Personal disposable income - Y
- Net liquid asset holdings - L

- Run three regressions – 2 with one explanatory variable, 1 with two explanatory variables.

standard errors - work out T-stats

$$(1)\ \hat{C} = \underset{(16)}{153} + \underset{(0.06)}{0.93}Y \quad R^2 = 0.93$$

$$(2)\ \hat{C} = \underset{(28)}{246} + \underset{(0.05)}{0.64}\,L \quad R^2 = 0.81$$

$$(3)\ \hat{C} = \underset{(61)}{84} + \underset{(0.53)}{0.78}Y + \underset{(0.37)}{0.43}L \quad R^2 = 0.95$$

- In equation (3) when Y and L are included together the standard errors (numbers in brackets) increase significantly.
- Not able reject null – so both explanatory variables are not significantly different from zero.
- But this appears to contradict the findings in equations (1) and (2).
- Results **<u>suggest</u>** - high collinearity between Y and L.

- Both C, Y and L - all increased steadily over last 50 years.

Why?

- All experienced strong upward trends.
- So likely Y and L highly collinear.
- So we are left uncertain as to the true importance either variable.

Detection of Multicollinearity

- No formal tests for multicollinearity exist.
- Simple partial (pairwise) correlation coefficients can be calculated to provide useful supporting evidence.
- High partial correlation coefficients between pairs/groups independent variables – indication of the potential existence of collinearity.
- Can also employ the correlation coefficient using a T-test to see if correlation between two explanatory variables is different from zero

- r_{12} = 0.87 and n=15

Hypothesis

- H_0: $\rho_{12} = 0$ and H_1: $\rho_{12} \neq 0$

Test Statistic

- TS = r_{12}/S_r where r_{12} is the sample correlation coefficient and

$$S_r = \sqrt{\frac{1 - r_{12}^2}{n-2}}$$

Critical Value

- T-test where α is the critical value and n-2 degrees of freedom

- So if S_r = 0.1367 then TS= 6.36 with CV = $\pm$ 2.16

Decision

- Reject H_0, of there being no correlation between variables 1 and 2.

- Running subsidiary regressions using each explanatory variable in turn:

- $X_2 = f(X_1, X_3, ...X_k)$
- $X_3 = f(X_1, X_2, ...X_k)$
- etc...

- For the auxiliary regressions inspect R^2 and F statistic. If high R^2 and regression is significant indication that multicollinearity might be a problem.

- Trial and error with excluding certain variables may reveal which combinations of variables are the cause of the multicollinearity, but very hit or miss to say the least!

Variance Inflation Factor (VIF)

- We can take the R^2s from the auxiliary regressions and use these to calculate VIFs.
- The VIF for any independent variable is a measure (albeit a little imprecise) of the degree of collinearity contributed by that variable.
- $VIF(X_i) = 1/(1-R^2_i)$
- As a rule of thumb if $VIF \geq 10$ which happens if $R^2 > 0.9$ then variable is said to be highly collinear.
- But high VIF need not be a clear indication of problem.

Solutions (all ad hoc)

- There are no definite solutions. Problem often amounts to poor (weak) data.

- So get more/better data (e.g. longer time series, or "pool" cross section and time series to get more independent variation in the variables).

- Re-specify the model to try and reduce the number of explanatory variables (but beware of omitted variable bias). There is no reason why we should do this and dropping explanatory variable - specification bias.

- Mis-specification of a model due to inclusion/exclusion of certain variables that results in a violation of theoretical principles.

- Specification bias is a serious problem for estimation reliability.

Example of Re-Specification

Impose constant returns to scale on the Cobb-Douglas function in order to reduce by one the number of variables:

$$\mathrm{Log}Q_t = a_0 + a_1 \mathrm{Log}K_t + a_2 \mathrm{Log}L_t + u_t \qquad (1)$$

Constant returns $\Rightarrow$ $a_1 + a_2 = 1$ so let $a_2 = 1 - a_1$ and rearrange (1) to obtain:

$$\mathrm{Log}\frac{Q_t}{L_t} = a_0 + a_1 \mathrm{Log}\frac{K_t}{L_t} + u_t$$

Improve the amount of independent variation in the data

Transforming a model into differences, e.g.:

$$\Delta Y_t = b_0 + b_1 \Delta X_{1t} + b_2 \Delta X_{2t} + u_t$$

$$\text{where} \quad \Delta Y_t = Y_t - Y_{t-1} \quad \text{etc.}$$

Transforming a model into differences and levels of the data, e.g.:

$$\Delta Y_t = b_0 + b_1 \Delta X_t + b_3 Y_{t-1} + b_4 X_{t-1} + u_t$$

Transform a model into logarithms

Do Nothing – Live with M/C

- If your t test estimates are just OK you will not make any inference errors even though you know that they are not as large as they ought to be.

- In practice Multicollinearity is a data problem, so acquiring better data is often the best solution, but can be difficult to achieve.

- Alternatively model transformations can be helpful

- In almost all applied research, we watch out for M/C and unless very strong we simply “live with it”…

Summary

- The distinction between perfect and imperfect but high multicollinearity
- The symptoms, detection and solutions for multicollinearity introduced
- Main message – much economic data and resulting model specification can be subject to multicollinearity – but we live with it, but always remain alert!
- There are various steps we can take to minimise
- But there is no definitive test to assess how serious the problem is.